

Zephyr Workspace Setup Guide

Setup steps for the latest Zephyr inside `myworkspace/` using Python 3.12.

Note: Python 3.12 is required. Zephyr's CMake modules call `find_package(Python3 3.12)` — using 3.9 or 3.10 will cause a CMake error at build time. Use the system `python3.12 (/usr/bin/python3.12)`.

Step 1 — Create a virtual environment with west

```
cd /home/jovin/workspace/My-Zephyr-project/myworkspace
python3.12 -m venv .venv
source .venv/bin/activate
pip install west
```

Step 2 — Initialize the Zephyr workspace

```
unset ZEPHYR_BASE # clear any existing ZEPHYR_BASE before init
west init .
west update
```

Downloads Zephyr kernel + all HAL/module dependencies (~1–2 GB, takes a few minutes).

Step 3 — Install Zephyr Python requirements

```
pip install -r zephyr/scripts/requirements.txt
```

Step 4 — Install the Zephyr SDK (ARM toolchain for nRF52840)

Check if the SDK is already installed first:

```
ls ~/zephyr-sdk-* 2>/dev/null && echo "SDK found" || echo "SDK not installed"
```

If not installed:

```
cd ~
wget https://github.com/zephyrproject-rtos/sdk-ng/releases/download/v1.0.1/zephyr-sdk-1.0.1_linux-x86_64.tar.xz
tar xf zephyr-sdk-1.0.1_linux-x86_64.tar.xz
cd zephyr-sdk-1.0.1
./setup.sh
```

Note: SDK 1.0.1 is required for Zephyr 4.x. SDK 0.16.8 and 0.17.x are incompatible — they cause a CMake “could not find Zephyr-sdk compatible with version 1.0” error.

Step 5 — Build a project

Run these commands at the start of every new terminal session:

```
source /home/jovin/workspace/My-Zephyr-project/myworkspace/.venv/bin/activate
export ZEPHYR_BASE=/home/jovin/workspace/My-Zephyr-project/myworkspace/zephyr
```

Then navigate to any project and build:

```
cd /home/jovin/workspace/My-Zephyr-project/src/2026-05-16_Introduction_to_Zephyr_RTOS
west build -b nrf52840dk/nrf52840
```

Flash to the board:

```
west flash
```

Quick reference — session setup (copy-paste)

```
source /home/jovin/workspace/My-Zephyr-project/myworkspace/.venv/  
bin/activate  
export ZEPHYR_BASE=/home/jovin/workspace/My-Zephyr-project/  
myworkspace/zephyr
```

Notes

- **Use python3.12** (/usr/bin/python3.12) — Zephyr's CMake requires Python ≥ 3.12 . Using 3.9 or 3.10 causes a CMake Could NOT find Python3 error at build time.
- ZEPHYR_BASE must **not** be set before running `west init` — unset it first if it points to ~/ncs/v2.4.2.
- The .venv folder is inside myworkspace/ which is excluded from git (.gitignore).
- If west is not found after activating the venv, re-run `pip install west` inside the activated environment.
- **SDK version 0.17.0** is required for Zephyr 4.x. SDK 0.16.8 only supports Zephyr up to 3.x.